



Grade 3

FAST Mathematics

Sample Test Materials

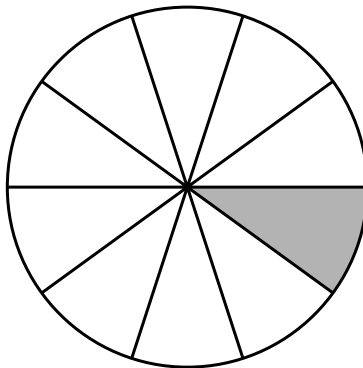
The purpose of these sample test materials is to orient teachers and students to the types of paper-based FAST Mathematics questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

Use the space in this Test and Response Book to do your work. Then, completely fill in the bubble beside the answer you choose. For some items, filling in more than one bubble may be required, so read each item carefully. If you change your answer, be sure to erase completely.

Some items will ask you to write a response in a shaded box or boxes. See the sample item below.

Sample Item:

A circle with one part shaded is shown.



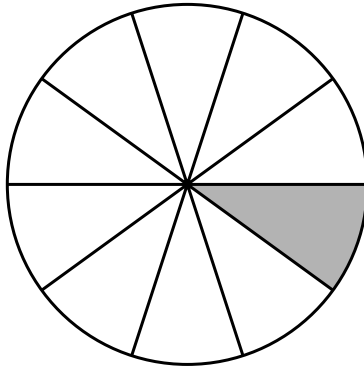
Create a fraction that represents the shaded part of the circle.

Write your response in the shaded box below.

	←	Write your response in the box.
--	---	--

Some items may have more than one box, so read each item carefully. Your answers for the items with response boxes may contain whole numbers or fractions.

1. A circle with one part shaded is shown.



Create a fraction that represents the shaded part of the circle.

Write your response in the shaded box below.

2. What is the expanded form of 7,403?

- Ⓐ $700 + 40 + 3$
- Ⓑ $7,000 + 40 + 3$
- Ⓒ $7,000 + 400 + 3$
- Ⓓ $7,000 + 400 + 30$

3. Ms. Devitt asks each of her 18 students to bring their favorite book to class.

- 6 students bring science fiction books.
- 3 students bring comic books.
- The rest of the students bring fantasy books.

Create a scaled picture graph to show the data. For each box, fill in the bubble before the correct number or picture.

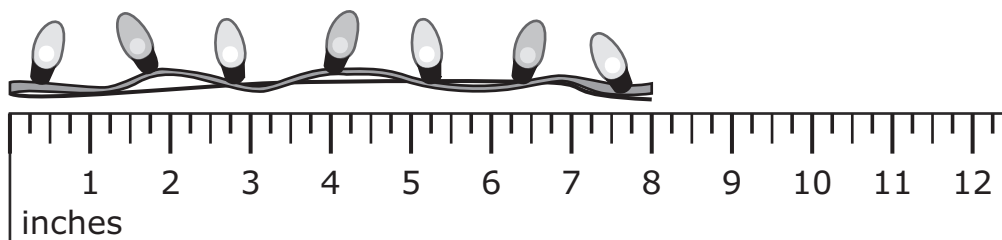
Key:  =

A	2
B	3

 students

Favorite Books	
Science fiction	A  B   C    D      
Comic	A  B   C    D      
Fantasy	A  B   C    D      

- 4.** Robert measures a string of lights, as shown.



How long, in inches, is the string of lights?

Write your response in the shaded box below.

5. This question has **two** parts.

An equation with an unknown whole number, u , is shown.

$$u \times 12 = 24$$

Part A

Which expression can be used to find the value of the unknown whole number, u ?

- Ⓐ $24 + 12$
- Ⓑ $24 - 12$
- Ⓒ 24×12
- Ⓓ $24 \div 12$

Part B

What is the value of the unknown whole number, u ?

Write your response in the shaded box below.

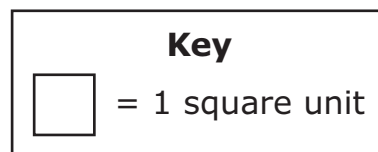
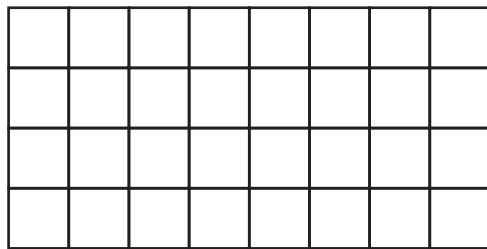
- 6.** A comparison with a missing number is shown.

$$\square > 7,652$$

Select all the numbers that could be the missing number.

- Ⓐ 7,526
- Ⓑ 7,652
- Ⓒ 7,752
- Ⓓ 7,662
- Ⓔ 7,625

7. A rectangle is tiled with unit squares, as shown.



Kenny says that he can find the area of the rectangle by counting all of the tiles.

Hank says there is another way to find the area.

Select a word and number to explain how Hank could find the area without counting the tiles.

For each box, fill in the bubble before the word or number that is correct.

Hank could

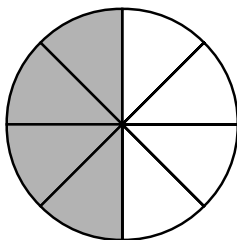
(A) add
(B) subtract
(C) multiply
(D) divide

 the side lengths of 4 and 8 to get an area of

(A) 2
(B) 4
(C) 12
(D) 32

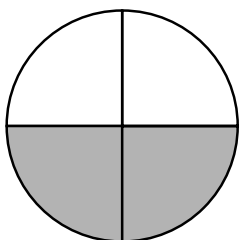
 square units.

- 8.** A fraction model is shown.

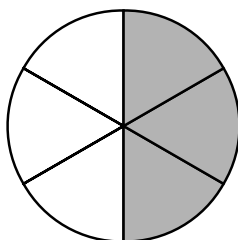


Select all the fraction models that are equivalent to the model shown.

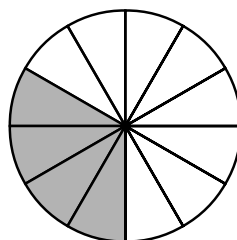
(A)



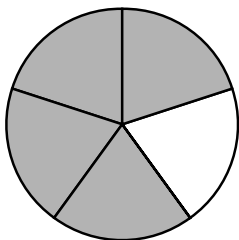
(C)



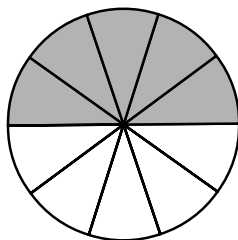
(E)



(B)



(D)



- 9.** Jamael does his math and science homework. He starts working on his homework at 3:57 p.m. and finishes at 4:32 p.m. He does not take a break.

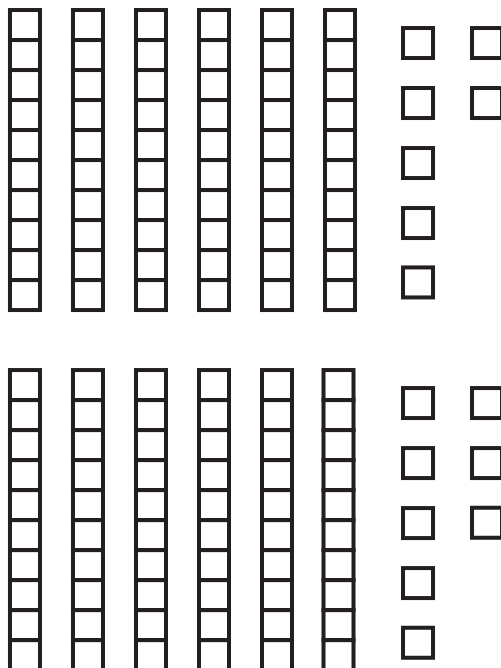
How many minutes could Jamael have worked on each subject?

Write your responses in the shaded boxes below.

Math:

Science:

10. A model of a number is shown.



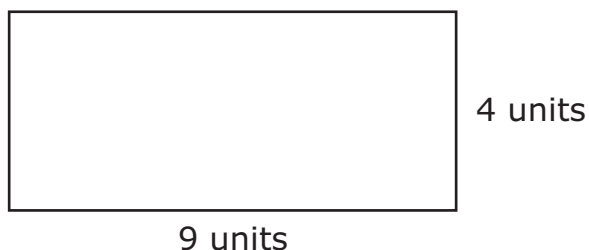
Complete the statement to describe whether the number is even or odd.
For each blank, fill in the bubble **before** the word that is correct.

The number is _____ [☐ A even ☐ B odd] because

it _____ [☐ A can ☐ B cannot] be split into two groups of equal size.

11. This question has **two** parts.

A rectangle is shown.



Part A

Fill in bubbles to select the expression that represents the area (in square units) and the perimeter (in units) of the rectangle.

	9 + 4	9 + 9 + 4 + 4	9 × 4
Area	(A)	(B)	(C)
Perimeter	(D)	(E)	(F)

Part B

Fill in bubbles to select the value that represents the area (in square units) and the perimeter (in units) of the rectangle.

	13	22	26	36
Area	(A)	(B)	(C)	(D)
Perimeter	(E)	(F)	(G)	(H)



Grade 3

FAST Mathematics

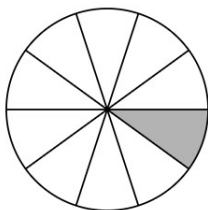
Sample Items

Answer Key

The Grade 3 FAST Mathematics Sample Items Answer Key provides the correct response(s) for each item on the sample test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test.

FAST Mathematics Sample Items Answer Key

1. A circle with one part shaded is shown.



Create a fraction that represents the shaded part of the circle.

$\frac{1}{10}$			
← → ↶ ↷ ✕			
1	2	3	
4	5	6	
7	8	9	
0	$\frac{\Box}{\Box}$		

Other correct responses: any equivalent fraction

FAST Mathematics Sample Items Answer Key

2. What is the expanded form of 7,403?

☐ A $700 + 40 + 3$

☐ B $7,000 + 40 + 3$

☒ C $7,000 + 400 + 3$

☐ D $7,000 + 400 + 30$

Option C: **This answer is correct.** The student correctly identified that the number has 7 thousands, 4 hundreds, and 3 ones.

FAST Mathematics Sample Items Answer Key

3. Ms. Devitt asks each of her 18 students to bring their favorite book to class.

- 6 students bring science fiction books.
- 3 students bring comic books.
- The rest of the students bring fantasy books.

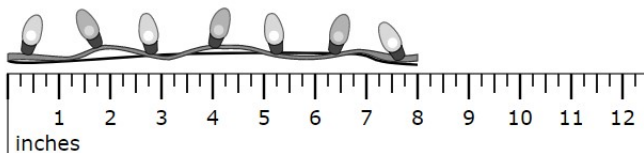
Create a scaled picture graph to show the data.

Key:  =  students

Favorite Books	
Science fiction	  2 
Comic	 1 
Fantasy	   3 

FAST Mathematics Sample Items Answer Key

4. Robert measures a string of lights, as shown.



How long, in inches, is the string of lights?

← → ↶ ↷ ✕

1	2	3
4	5	6
7	8	9
0	$\frac{\Box}{\Box}$	

Other correct responses: any equivalent value

FAST Mathematics Sample Items Answer Key

5. This question has **two** parts.

An equation with an unknown whole number, u , is shown.

$$u \times 12 = 24$$

Part A

Which expression can be used to find the value of the unknown whole number, u ?

☐ (A) $24 + 12$

☐ (B) $24 - 12$

☐ (C) 24×12

☒ (D) $24 \div 12$

Part B

What is the value of the unknown whole number, u ?

2

← → ↶ ↷ ✕			
1	2	3	
4	5	6	
7	8	9	
0	$\frac{\Box}{\Box}$		

Option D: **This answer is correct.** The student selected the related division expression.

FAST Mathematics Sample Items Answer Key

6. A comparison with a missing number is shown.

$$\square > 7,652$$

Select all the numbers that could be the missing number.

☐ 7,526

☐ 7,652

☒ 7,752

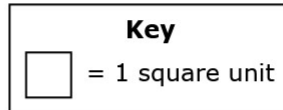
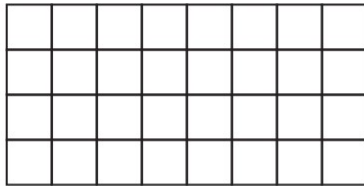
☒ 7,662

☐ 7,625

Option C: **This answer is correct.** The student identified the hundreds digit of 7 as being greater than a hundreds digit of 6.

Option D: **This answer is correct.** The student identified the thousands and hundreds digits equal to the given number and a tens digit greater than the 5 in 7,652.

7. A rectangle is tiled with unit squares, as shown.



Kenny says that he can find the area of the rectangle by counting all of the tiles.

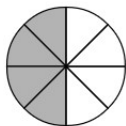
Hank says there is another way to find the area.

Select a word and number to explain how Hank could find the area without counting the tiles.

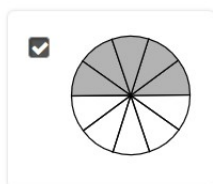
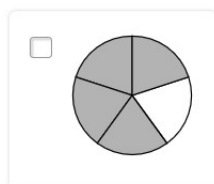
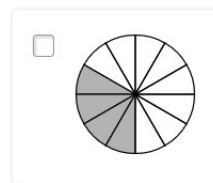
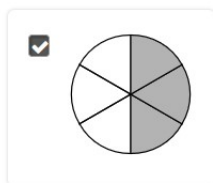
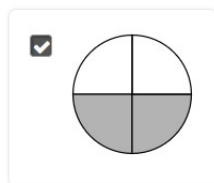
Hank could the side lengths of 4 and 8 to get an area of square units.

FAST Mathematics Sample Items Answer Key

8. A fraction model is shown.



Select all the fraction models that are equivalent to the model shown.



Option A: **This answer is correct.** The student determined that this model is equivalent to the given model because each model has the same fraction of the whole shaded.

Option C: **This answer is correct.** The student determined that this model is equivalent to the given model because each model has the same fraction of the whole shaded.

Option D: **This answer is correct.** The student determined that this model is equivalent to the given model because each model has the same fraction of the whole shaded.

FAST Mathematics Sample Items Answer Key

9. Jamael does his math and science homework. He starts working on his homework at 3:57 p.m. and finishes at 4:32 p.m. He does not take a break.

How many minutes could Jamael have worked on each subject?

Math:

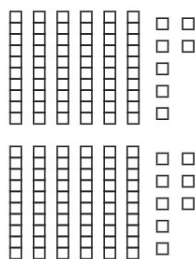
Science:

←	→	↶	↷	✖
1	2	3		
4	5	6		
7	8	9		
0	$\frac{\Box}{\Box}$			

Other correct responses: any two amounts of time whose sum is 35 minutes where neither value is equal to 0

FAST Mathematics Sample Items Answer Key

- 10.** A model of a number is shown.



Complete the statement to describe whether the number is even or odd.

The number is because it be split into two groups of equal size.

FAST Mathematics Sample Items Answer Key

11. This question has **two** parts.

A rectangle is shown.



Part A

Select the expression that represents the area (in square units) and the perimeter (in units) of the rectangle.

	$9 + 4$	$9 + 9 + 4 + 4$	9×4
Area	☐	☐	☒
Perimeter	☐	☒	☐

Part B

Select the value that represents the area (in square units) and the perimeter (in units) of the rectangle.

	13	22	26	36
Area	☐	☐	☐	☒
Perimeter	☐	☐	☒	☐

FAST Mathematics Sample Items Answer Key

- 12.** The table shows the number of raffle tickets students in grades 3, 4, and 5 sell.

Grade	Number of Tickets
3	24
4	15
5	27

Click on full or half tickets to create a pictograph of the data.

Raffle Tickets Students Sell

Grade	Number of Tickets
3	 
4	  
5	  

Each  represents 6 tickets.

FAST Mathematics Sample Items Answer Key

- 13.** An equation is shown.

$$72 \div 6 = 4 \times 3$$

Enter numbers and select a word to complete the statement.

Since $72 \div 6 =$ and $4 \times 3 =$, the equation is .

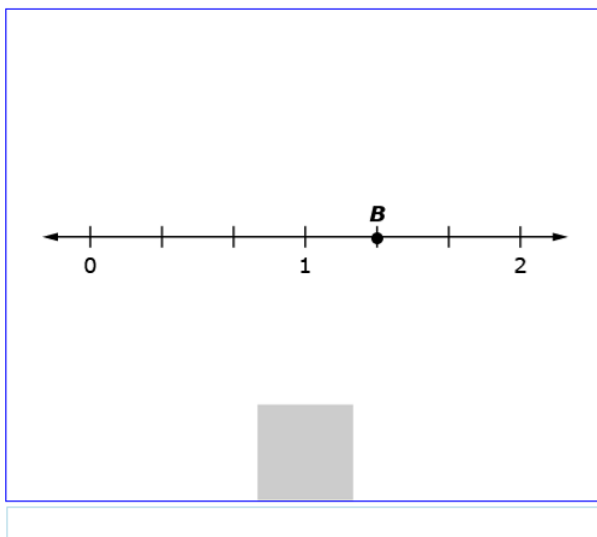
1	2	3		
4	5	6		
7	8	9		
0				

FAST Mathematics Sample Items Answer Key

- 14.** An expression is shown.

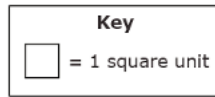
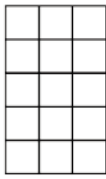
$$\frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$$

Drag point B to the number line to represent the value of the expression.



FAST Mathematics Sample Items Answer Key

15. A rectangle is shown.



Create a multiplication equation that shows the area, in square units, of the rectangle.

$3 \times 5 = 15$

← → ↶ ↷ ✕

1	2	3	+	-	×	÷
4	5	6	<	=	>	
7	8	9	()			
0	$\frac{\Box}{\Box}$					

Other correct responses: any equivalent multiplication equation that is equal to 15 on both sides and contains 15

FAST Mathematics Sample Items Answer Key

16. This question has **two** parts.

Angela makes an error when finding the product of 3 and 42. Her work is shown.

Part A

Click on the part of the work where Angela's error first appears.

Given	3×42
Step 1	$3 \times (4 + 2)$
Step 2	$(3 \times 4) + (3 \times 2)$

Part B

What is the product of 3 and 42?

←

→

↶

↷

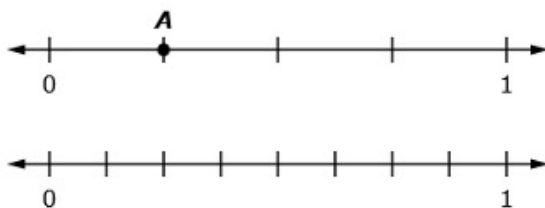
✖

1	2	3
4	5	6
7	8	9
0	$\frac{\Box}{\Box}$	

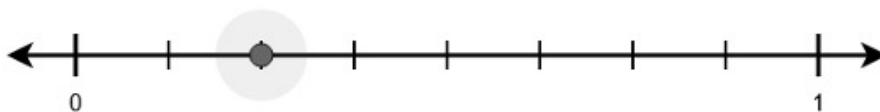
Other correct responses: for Part B, any equivalent value

FAST Mathematics Sample Items Answer Key

- 17.** Two number lines representing the same whole are shown. Point *A* is plotted to represent $\frac{1}{4}$.



Click on the number line below to plot a point that represents a fraction equivalent to $\frac{1}{4}$.



FAST Mathematics Sample Items Answer Key

- 18.** Create and shade a model to represent the fraction $\frac{1}{6}$.

Divide the model into equal parts using the – or + button. Then, select each part that should be shaded.

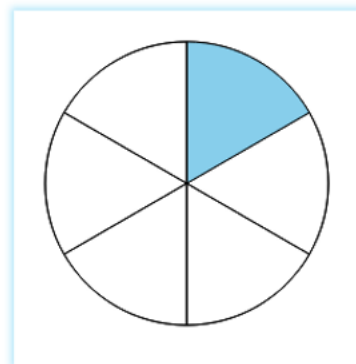
Number of Parts



6

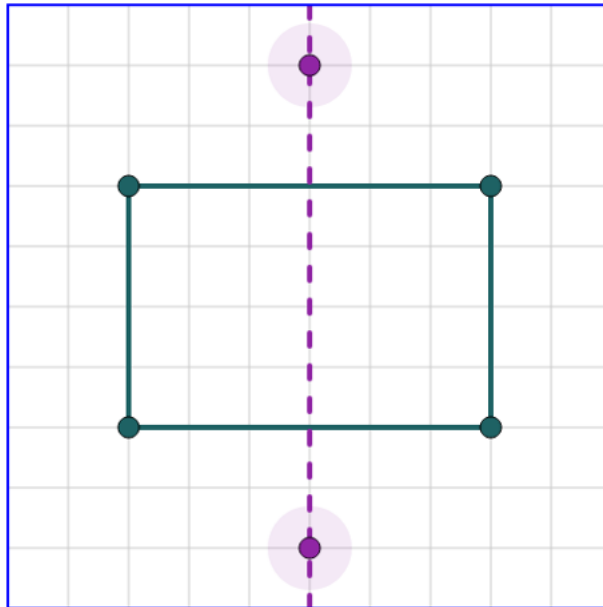


Reset

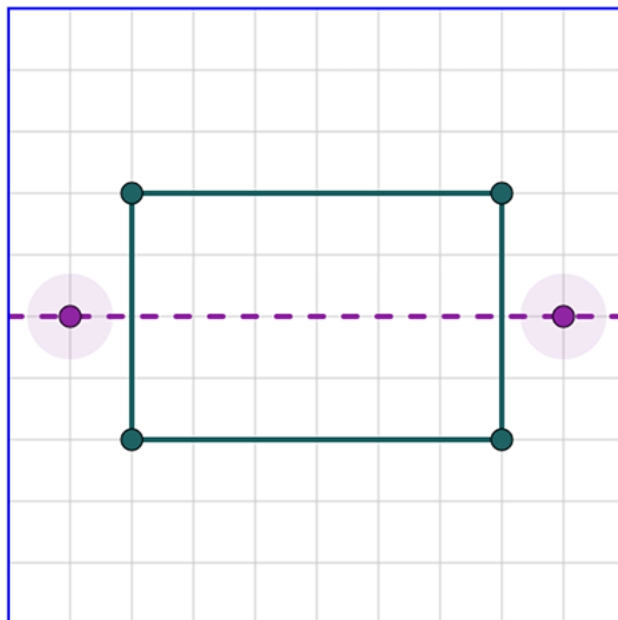


Note: or any model that represents $\frac{1}{6}$

- 19.** Click on the grid to draw a line of symmetry for the shape.

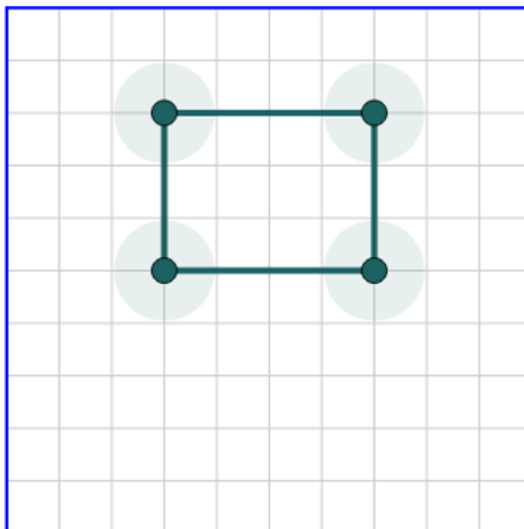


or



FAST Mathematics Sample Items Answer Key

20. Click on the grid to draw a quadrilateral with 4 right angles.



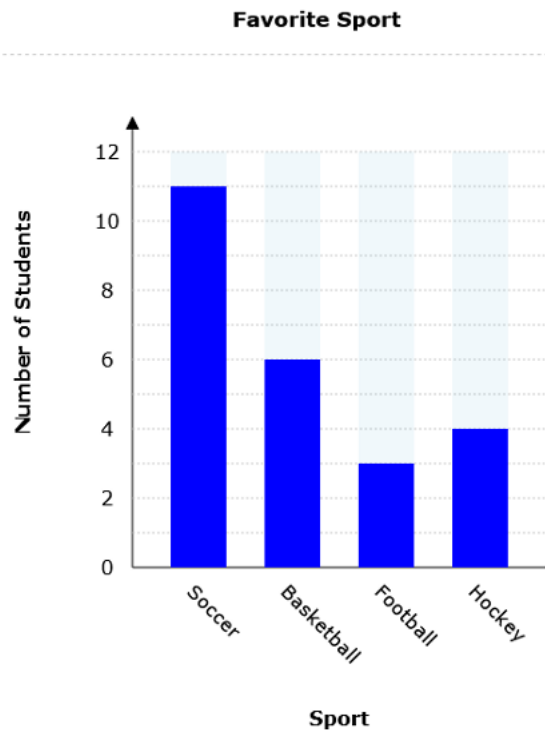
Other correct responses: any quadrilateral with four right angles

FAST Mathematics Sample Items Answer Key

- 21.** A teacher asks her students what their favorite sports are. The table shows the results.

Favorite Sport	
Sport	Number of Students
Soccer	11
Basketball	6
Football	3
Hockey	4

Click above each sport to create a bar graph of the data.



FAST Mathematics Sample Items Answer Key

- 22.** Julia measures the lengths of some ribbons to the nearest inch. Her measurements are shown.

4, 7, 8, 7, 4, 9, 8, 5, 7

Create a line plot for Julia's data.

Ribbon Lengths

